Unit 6

Campbell Biology, AP Edition, 12e

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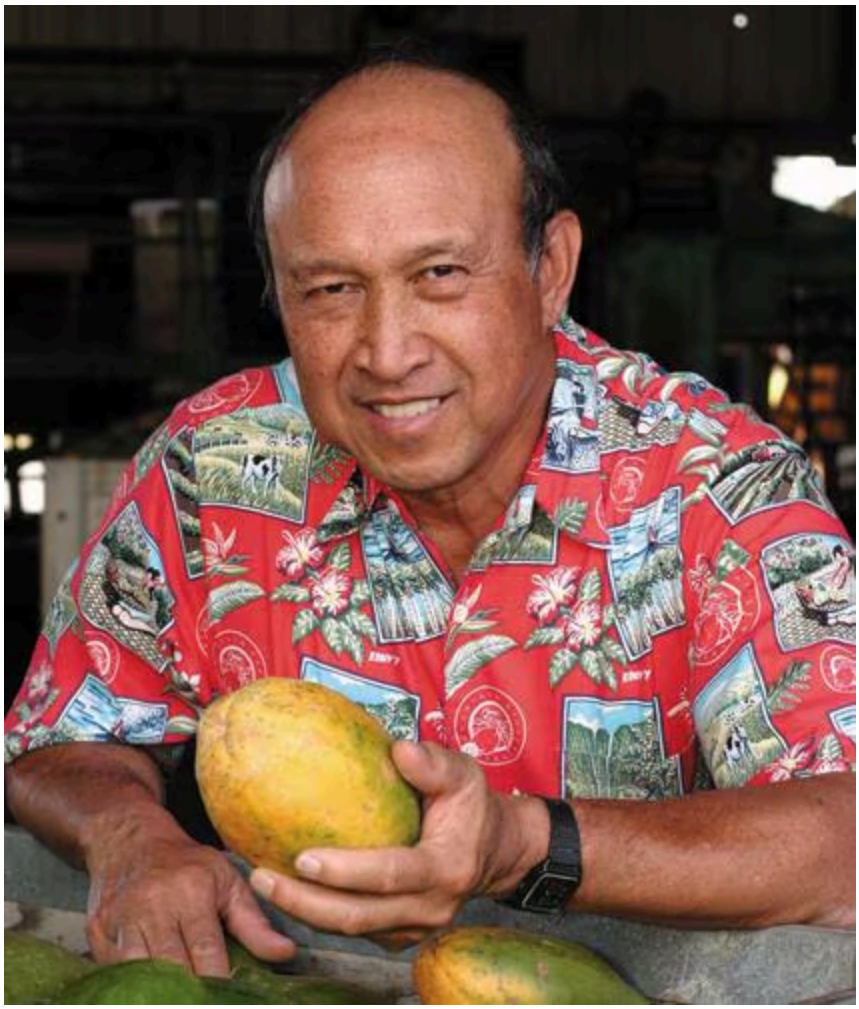
Plant Form and Function

Dennis Gonsalves



Meet the man who saved the papaya industry in Hawaii, Dennis Gonsalves. Dr. Gonsalves received a B.S. in Horticulture and a M.S. in Plant Pathology from the University of Hawaii, Manoa, and a Ph.D. in Plant Pathology from the University of California, Davis. He became an Associate Professor of plant pathology at the University of Florida and later a Liberty Hyde Bailey Professor at Cornell University. He and his colleagues created virus-resistant papaya plants that rescued the papaya-growing industry in Hawaii from the devastating effects of the ringspot virus. He then served as the Director of the Agricultural Research

Center in Hilo, Hawaii, until he retired in 2012. He received the Humboldt Prize for renowned scientific research in 1992 and the Lee Hutchison Award for accomplishments in research, mentoring, and outreach to developing countries.

An Interview with: Dennis Gonsalves

How did you get interested in plant biology?

I was born on a sugar plantation in Kohala, Hawaii, so I grew up in an agricultural setting. My dad was a laborer on the Kohala Sugar Plantation. I attended the University of Hawaii to study agriculture engineering because my goal was to get a degree and then work on the sugar plantations as an agriculture engineer. But when I was a sophomore, the sugar plantations were closing in Hawaii, and the program was disbanded. I took up horticulture as an alternative. After graduation, I moved to Kauai, where there was a University of Hawaii experiment station where I worked as a technician in a plant pathology lab. There, I was a given a small project of my own to work on, and I fell in love with research.

Tell us about your experiences in graduate school.

I went to UC Davis for my Ph.D. I understood the material. I could do the research. But when it came to giving lectures, I was hampered by a stuttering problem. A few months before I got my Ph.D., I applied for a job at the University of Florida and was invited for an interview. And then I thought, “I have to interview. I am a poor speaker. I don’t stand a chance if I do a poor job.” I was sick with stress. I practiced and practiced, and then it dawned on me, “I know more about this topic than my professors do. I’m the expert.” So, I flew to Florida, and I will never forget that day: I got up and gave my talk, and I have not stuttered since!

What is the ringspot virus and how did you become interested in it?

The virus produces a ring spot on the fruit, but more importantly, the leaves turn yellow, the growth gets stunted, and often the plant won’t produce fruit. It is rapidly transmitted by aphids, small sap-sucking insects. An aphid can land on an infected papaya, feed for less than a minute, then jump to a healthy papaya plant and infect it. In 1978, I returned to Hawaii for a short vacation and I met a colleague who told me, “Dennis, there is this virus disease on the Big Island where they grow 95% of the papaya, and it’s close to the papaya-growing areas. People are trying to prevent it from spreading.” So, when I returned to Cornell, I tried to figure out how to control the papaya ringspot virus.

Top: A papaya tree infected with ringspot virus. Inset: An infected papaya fruit. Bottom: A "Rainbow" papaya tree that is resistant to ringspot virus.

What was your strategy in fighting the disease?

In our initial attempt, we used artificial selection to isolate a mild strain of papaya ringspot virus to protect plants against infection, but it didn't grow well under field conditions. I have learned, however, as a scientist, sometimes you have to change directions. I was a classical virologist, but I decided to retrain myself as a molecular biologist. Research by others with tobacco had established that certain viral genes, if introduced into the host plant genome, induce resistance. Other colleagues at Cornell had invented a "gene gun" that uses fine, DNA-coated tungsten beads to literally shoot DNA into cells. We shot tungsten beads coated with DNA of the coat protein gene of the ringspot virus into papaya embryos growing in culture. Some of these genetically modified embryos could be grown into plants. When I inoculated them with the virus, one line was completely resistant; it became a parent of the resistant "Rainbow" variety. "Rainbow" was developed just in time because the virus had spread into the main papaya-growing areas and was decimating the papaya crop.

> ***"As a scientist, sometimes you have to change directions."***

Have foes of genetically modified foods opposed your research?

In 1994, we teamed up with Thai researchers to repeat our success using a large-fruited papaya variety popular in Thailand. Ten years later, we were successful, but opponents of genetic engineering stormed the fences surrounding our field trials and destroyed many of the fruit. That transgenic papaya has never seen the light of day. Although I've had a pretty good career, that was a heartbreaking thing for me.